

Proof-Carrying Exact Quantum Bounds for the I_{3322} Bell Inequality

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Abstract

We give a proof-carrying exact-arithmetic enclosure for the tensor-product and commuting-operator quantum values of the I_{3322} Bell inequality. A rational zero-residual Gram certificate on the full word space of length at most four proves

$$Q_c(I_{3322}) \leq U_4 := \frac{501750771}{2000000000} = 0.2508753855.$$

The certificate is reduced by an exact signed D_4 symmetry into five positive-definite rational blocks of dimensions 31, 30, 26, 35, and 61; the last block occurs twice in the 244-dimensional word representation. An explicit local-dimension-499 tensor strategy, reconstructed from rational rank-one projector blocks and a rational Schmidt vector, has exact value

$$L_{499} = 0.2508753845139766 \dots$$

Consequently

$$L_{499} \leq Q_t(I_{3322}) \leq Q_c(I_{3322}) \leq U_4, \quad U_4 - L_{499} = 9.860234645173508 \dots \times 10^{-10}.$$

A compact statement printable in full is

$$\frac{25087538451}{100000000000} < Q_t(I_{3322}) \leq Q_c(I_{3322}) \leq \frac{501750771}{20000000000},$$

whose exact width is $99/100000000000 = 9.9 \times 10^{-10}$. This is about a 265-fold reduction of the previous exact enclosure. We derive the reduced dimension-499 formula from the explicit projectors, clarify the convention behind the exact length-one value $3/8$, and give a precise verification contract and trust map. The release uses canonical non-executable JSON, explicit exception-based checks that remain active under optimized Python, two separately implemented standard-library verifiers, direct reconstruction of the high-dimensional strategy, mutation tests, continuous integration, and a release-wide integrity manifest. No floating-point output enters an acceptance decision.

1 Introduction

The I_{3322} inequality is the simplest tight bipartite Bell inequality with three binary measurements per party that is inequivalent to CHSH [2]. It is a standard test case for semidefinite outer approximations of quantum correlations and for finite-dimensional variational searches. Numerical work places its quantum value near 0.250875384514 in the Collins–Gisin normalization and indicates unusually persistent dimension dependence [4, 6, 10].

High-precision numerical evidence and an exact theorem have different logical status. A floating-point SDP matrix can lose positive semidefiniteness after rounding, an approximate sum-of-squares identity can conceal a small residual, and rounded measurement matrices need not satisfy projector or involution relations exactly. The aim here is therefore not to discover additional decimal digits. It is to convert a sharp numerical picture into a finite proof object whose acceptance requires only exact integer and rational arithmetic.

The principal results are:

- (i) a full length-four, zero-residual rational upper certificate

$$Q_c(I_{3322}) \leq 501750771/2000000000;$$

- (ii) an explicit exact dimension-499 tensor strategy with value L_{499} and a direct derivation of its tridiagonal Rayleigh quotient;
- (iii) an exact enclosure of width $9.860234645 \dots \times 10^{-10}$, hence the same upper bound on any tensor-versus-commuting gap witnessed by this functional;
- (iv) exact upper-certificate progression at the bare length-one, $1+AB$, full length-two, full length-three, and full length-four word sets, together with exact strategies in dimensions 12, 16, and 499;
- (v) a hardened proof release with two source-independent verification paths, direct projector reconstruction, optimized-mode tests, mutation tests, and an explicit theorem-dependency map.

The result does *not* identify the exact Tsirelson bound, prove qubit optimality, settle finite-dimensional attainment, or prove a tensor–commuting separation. Its main contribution is exact certification: the numerical optimizer is removed from the trusted computing base.

2 Bell functional and quantum models

Alice and Bob each have three binary observables. In a commuting-operator model, A_i and B_j are self-adjoint involutions on a Hilbert space and satisfy $[A_i, B_j] = 0$. Tensor-product models are the special case $A_i = \tilde{A}_i \otimes \mathbb{I}$ and $B_j = \mathbb{I} \otimes \tilde{B}_j$. We denote the corresponding suprema by Q_c and Q_t , respectively; always $Q_t \leq Q_c$.

In the Collins–Gisin probability normalization,

$$\begin{aligned} I_{3322} = & P(A_1 B_1) + P(A_1 B_2) + P(A_1 B_3) + P(A_2 B_1) + P(A_2 B_2) - P(A_2 B_3) \\ & + P(A_3 B_1) - P(A_3 B_2) - P(A_1) - 2P(B_1) - P(B_2). \end{aligned} \quad (1)$$

For ± 1 observables this is equivalent to

$$4I_{3322} = \langle \mathcal{B}_{3322} \rangle - 4, \quad (2)$$

where

$$\begin{aligned} \mathcal{B}_{3322} = & A_1 + A_2 - B_1 - B_2 + A_1(B_1 + B_2 + B_3) \\ & + A_2(B_1 + B_2 - B_3) + A_3(B_1 - B_2). \end{aligned} \quad (3)$$

The classical bound is $I_{3322} \leq 0$.

Let $G_3 = \mathbb{Z}_2 * \mathbb{Z}_2 * \mathbb{Z}_2$ and $G = G_3 \times G_3$. A normal-form element is a pair (u, v) of reduced single-party words. Let $\mathcal{W}_k$ contain all words of total length at most k . The cardinalities relevant here are

$$|\mathcal{W}_1| = 7, \quad |\mathcal{W}_2| = 28, \quad |\mathcal{W}_3| = 88, \quad |\mathcal{W}_4| = 244.$$

The intermediate set

$$\mathcal{W}_{1+AB} = \{\mathbb{I}, A_i, B_j, A_i B_j : 1 \leq i, j \leq 3\}$$

has cardinality 16.

3 Exact certificate calculus

Let $W = (w_1, \dots, w_N)^\top$ be a word vector and let $Y \in \mathbb{Q}^{N \times N}$ be symmetric.

Proposition 3.1 (Zero-residual upper certificate). *If $Y \succeq 0$ and, exactly in $\mathbb{Q}[G]$,*

$$\beta\mathbb{K} - \mathcal{B}_{3322} = W^*YW, \quad (4)$$

then

$$Q_c(I_{3322}) \leq \frac{\beta - 4}{4}.$$

Proof. Every commuting representation sends W^*YW to a positive operator. Hence $\langle \mathcal{B}_{3322} \rangle \leq \beta$, and (2) gives the claim. $\square$

Lemma 3.2 (Residual absorption). *If $Y \succeq 0$ and*

$$\beta\mathbb{K} - \mathcal{B}_{3322} - W^*YW = \sum_g r_g g,$$

then $\langle \mathcal{B}_{3322} \rangle \leq \beta + \sum_g |r_g|$.

Proof. Every group element is represented by a unitary, so any state ϕ obeys $|\phi(\sum_g r_g g)| \leq \sum_g |r_g|$. Combine this with positivity of W^*YW . $\square$

Proposition 3.3 (Exact tensor lower certificate). *Let six rational matrices define exact finite-dimensional self-adjoint involutions and let $0 \neq \psi$ be a rational bipartite vector. Then the exact Rayleigh quotient of $\mathcal{B}_{3322}$ on ψ , normalized through (2), is a lower bound on $Q_t(I_{3322})$.*

These statements are elementary; the substantive work is to produce sharp proof objects for which positivity, the group-algebra identity, and the model relations can all be checked exactly.

4 The bare length-one relaxation

The following identity gives the exact dual value of the bare word-length-one involution relaxation.

Proposition 4.1 (Closed-form sum of squares). *In $\mathbb{C}[G]$,*

$$\begin{aligned} \frac{11}{2}\mathbb{K} - \mathcal{B}_{3322} = \frac{1}{2}[(A_1 + A_2 - B_1 - B_2 - \mathbb{K})^2 \\ + (A_1 - A_2 - B_3)^2 + (A_3 - B_1 + B_2)^2]. \end{aligned} \quad (5)$$

Proof. Put $C = A_1 + A_2$, $D = A_1 - A_2$, $S = B_1 + B_2$, and $T = B_1 - B_2$. Expanding and using $C^2 + D^2 = S^2 + T^2 = 4\mathbb{K}$ gives $11\mathbb{K} - 2\mathcal{B}_{3322}$. $\square$

In the word order $(\mathbb{K}, B_1, B_2, B_3, A_1, A_2, A_3)$, the release contains the rational moment matrix

$$\Gamma_1 = \begin{pmatrix} 1 & -1/4 & -1/4 & 0 & 1/4 & 1/4 & 0 \\ -1/4 & 1 & 1/2 & 0 & 5/8 & 5/8 & 1/2 \\ -1/4 & 1/2 & 1 & 0 & 5/8 & 5/8 & -1/2 \\ 0 & 0 & 0 & 1 & 1/2 & -1/2 & 0 \\ 1/4 & 5/8 & 5/8 & 1/2 & 1 & 1/2 & 0 \\ 1/4 & 5/8 & 5/8 & -1/2 & 1/2 & 1 & 0 \\ 0 & 1/2 & -1/2 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

It is word-consistent, positive semidefinite of rank four, and has objective $11/2$.

Theorem 4.2 (Exact bare length-one value). *The word-length-one moment relaxation containing only positive semidefiniteness and the involution/group-word consistency relations has exact value*

$$I_{3322}^{(1, \text{bare})} = \frac{3}{8}.$$

Proof. Proposition 4.1 is dual feasible and Γ_1 is primal feasible with the same objective. $\square$

Remark 4.3 (Convention). *Some presentations of the first NPA level append explicit nonnegativity constraints for measurement-event probabilities. Those inequalities are not automatic in the bare length-one involution moment matrix. The value $3/8$ proved here refers precisely to the bare relaxation just defined; it should not be identified with a strengthened probability-positive Q_1 convention.*

5 The full length-four upper certificate

The principal new upper certificate uses the complete 244-word vector W_4 but exploits an exact symmetry of the Bell polynomial.

5.1 Signed D_4 symmetry

Order the six generators as $(A_1, A_2, A_3, B_1, B_2, B_3)$. Define signed substitutions

$$\begin{aligned} r &= (-B_1, -B_2, B_3, -A_2, -A_1, -A_3), \\ s &= (A_1, A_2, -A_3, B_2, B_1, B_3). \end{aligned}$$

Direct substitution shows that both preserve $\mathcal{B}_{3322}$, and they obey

$$r^4 = s^2 = \mathbb{I}, \quad srs = r^{-1}.$$

They therefore generate a signed action of D_4 on $\mathcal{W}_4$.

The real word representation decomposes into the four one-dimensional characters and the unique real two-dimensional irreducible representation. The corresponding multiplicities are

$$31, \quad 30, \quad 26, \quad 35, \quad 61,$$

so that

$$31 + 30 + 26 + 35 + 2 \cdot 61 = 244.$$

The release constructs canonical integer basis matrices $U_{++}, U_{+-}, U_{-+}, U_{--}, U_+, U_-$ by exact character projection and deterministic modular pivot selection. Every invariant symmetric Gram matrix in the chosen commutant section has the form

$$\begin{aligned} Y &= U_{++}Z_{++}U_{++}^\top + U_{+-}Z_{+-}U_{+-}^\top + U_{-+}Z_{-+}U_{-+}^\top + U_{--}Z_{--}U_{--}^\top \\ &\quad + U_+Z_\rho U_+^\top + U_-Z_\rho U_-^\top. \end{aligned} \tag{6}$$

Thus positivity reduces to five rational blocks rather than one 244×244 matrix.

5.2 Exact certificate

The headline proof object stores five symmetric rational blocks

$$Z_{++} \in \mathbb{Q}^{31 \times 31}, \quad Z_{+-} \in \mathbb{Q}^{30 \times 30}, \quad Z_{-+} \in \mathbb{Q}^{26 \times 26}, \quad Z_{--} \in \mathbb{Q}^{35 \times 35}, \quad Z_\rho \in \mathbb{Q}^{61 \times 61}.$$

Each block is represented by an integer matrix and a positive common denominator.

Theorem 5.1 (Full length-four rational upper bound). *The blocks in the release are positive definite and reconstruct, through (6), an exact identity*

$$\frac{2501750771}{5000000000} \mathbb{I} - \mathcal{B}_{3322} = W_4^* Y W_4.$$

Consequently

$$Q_c(I_{3322}) \leq U_4 := \frac{501750771}{2000000000} = 0.2508753855.$$

Proof. The primary verifier proves positivity of every integer block by fraction-free Bareiss elimination and the Sylvester criterion. It independently regenerates the D_4 bases, reconstructs Y , reduces every product $w_i^* w_j$ to group normal form, and verifies every coefficient in the displayed identity exactly. Proposition 3.1 applies. The second verifier regenerates the symmetry bases in separately written code, proves positivity by exact rational $LDL^\top$ elimination, and accumulates group coefficients directly from the block columns without first forming Y . $\square$

The earlier exact certificates are retained to show the hierarchy progression:

$$\begin{aligned} Q_c(I_{3322}) &\leq \frac{1105981489019}{4398046511104} = 0.2514710761304\dots & (1+AB), \\ Q_c(I_{3322}) &\leq \frac{1103645539239}{4398046511104} = 0.2509399426433\dots & (\text{full length } 2), \\ Q_c(I_{3322}) &\leq \frac{68960172491}{274877906944} = 0.2508756460556469\dots & (\text{full length } 3), \\ Q_c(I_{3322}) &\leq \frac{501750771}{2000000000} = 0.2508753855 & (\text{full length } 4). \end{aligned}$$

The length-four statement is a certified feasible upper bound; no claim is made that it is the exact optimum of the length-four SDP.

6 Exact tensor strategies

6.1 Dense dimensions 12 and 16

The release contains six rational symmetric involutions and a rational bipartite state in each of local dimensions 12 and 16. Exact contraction gives

$$\begin{aligned} L_{12} &= 0.2500643651906238\dots, \\ L_{16} &= 0.2504433069760140\dots. \end{aligned}$$

In particular, dimension 12 is sufficient to exceed $1/4$. This statement does not prove that dimension 12 is necessary.

6.2 Rational block projectors in dimension 499

Let $n = 499$, set $c_0 = 1$ and $c_n = -1$, and let the certificate contain rational pairs (c_i, s_i) for $1 \leq i < n$ satisfying

$$c_i^2 + s_i^2 = 1, \quad s_i > 0.$$

Define rank-one projector blocks

$$\begin{aligned} R_i^\pm &= \frac{1}{2} \begin{pmatrix} 1 - c_i & \pm s_i \\ \pm s_i & 1 + c_i \end{pmatrix}, \\ S_i^\pm &= \frac{1}{2} \begin{pmatrix} 1 + c_i & \pm s_i \\ \pm s_i & 1 - c_i \end{pmatrix}, \\ J &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \end{aligned}$$

Their traces are one and their determinants vanish exactly, so they are orthogonal rank-one projectors.

Following the block pattern of Pál and Vértesi [6], define six projectors $\hat{A}_1, \hat{A}_2, \hat{A}_3, \hat{B}_1, \hat{B}_2, \hat{B}_3$ on the standard basis $e_1, \dots, e_n$ as follows:

- $\hat{A}_2$ has the singleton $|e_1\rangle\langle e_1|$ and blocks R_i^+ on (e_i, e_{i+1}) for even i ; $\hat{A}_1$ uses R_i^- on the same blocks.

- $\hat{B}_2$ has blocks S_i^+ on (e_i, e_{i+1}) for odd $i < n$ and a zero final singleton; $\hat{B}_1$ uses S_i^- .
- $\hat{A}_3$ has J on (e_i, e_{i+1}) for odd $i < n$ and the final singleton $|e_n\rangle\langle e_n|$.
- $\hat{B}_3$ has the first singleton $|e_1\rangle\langle e_1|$ and J on (e_i, e_{i+1}) for even i .

The measurement labeling used in (1) is

$$(P_{A_1}, P_{A_2}, P_{A_3}) = (\hat{A}_2, \hat{A}_1, \hat{A}_3), \quad (P_{B_1}, P_{B_2}, P_{B_3}) = (\hat{B}_2, \hat{B}_1, \hat{B}_3). \quad (7)$$

The associated ± 1 observables are $2P - \mathbb{I}$.

Let $\lambda_1, \dots, \lambda_n \in \mathbb{Q}$ be the stored nonzero Schmidt vector and put

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^n \lambda_i |ii\rangle, \quad N = \sum_{i=1}^n \lambda_i^2.$$

For real symmetric projectors P, Q ,

$$\langle\psi|P \otimes Q|\psi\rangle = \frac{1}{N} \sum_{i,j=1}^n \lambda_i \lambda_j P_{ij} Q_{ij}, \quad \langle\psi|P \otimes \mathbb{I}|\psi\rangle = \frac{1}{N} \sum_i \lambda_i^2 P_{ii}. \quad (8)$$

The normalization square root therefore cancels completely.

6.3 Derivation of the reduced tridiagonal formula

Because every projector is block diagonal with blocks on adjacent coordinates, (8) contains only diagonal terms λ_i^2 and nearest-neighbor terms $\lambda_i \lambda_{i+1}$. Substitute the projectors above into (1) and use the relabeling (7). The opposite signs in $R_i^\pm$ and $S_i^\pm$ cancel unwanted off-diagonal contributions and leave, for each $1 \leq i < n$, exactly

$$s_i \lambda_i \lambda_{i+1}.$$

Collecting all diagonal contributions gives

$$d_i = c_{i-1} c_i + \frac{c_{i-1} - c_i}{2} - 1 + \frac{c_n + 1}{2} \delta_{i,n}. \quad (9)$$

Since $c_n = -1$, the displayed boundary term vanishes here, but retaining it records the block-end convention explicitly. Hence the original probability expression is exactly

$$I_{3322}(\psi) = \frac{\sum_{i=1}^n d_i \lambda_i^2 + \sum_{i=1}^{n-1} s_i \lambda_i \lambda_{i+1}}{\sum_{i=1}^n \lambda_i^2}. \quad (10)$$

This is an algebraic identity, not a numerical approximation.

Theorem 6.1 (Exact dimension-499 lower strategy). *The released rational parameters define six exact projectors through the construction above. Their direct evaluation in (1) equals the reduced expression (10) and has exact value L_{499} recorded in `results.json`, with*

$$L_{499} = 0.2508753845139766 \dots > \frac{25087538451}{100000000000}.$$

Therefore $L_{499} \leq Q_t(I_{3322})$.

Proof. The primary verifier checks all 498 unit-circle identities, reconstructs every sparse projector, verifies symmetry and $P^2 = P$ exactly, checks that the Schmidt vector is nonzero, and evaluates (1) by direct rational contraction. It independently evaluates (10) and requires exact equality. The second verifier separately evaluates the reduced expression. Proposition 3.3 applies. $\square$

7 Certified enclosure and physical consequences

Theorem 7.1 (Main exact enclosure). *For the I_{3322} Bell inequality,*

$$L_{499} \leq Q_t(I_{3322}) \leq Q_c(I_{3322}) \leq \frac{501750771}{20000000000}.$$

The exact rational width is stored in `results.json` and satisfies

$$U_4 - L_{499} = 9.860234645173508 \dots \times 10^{-10} < 10^{-9}.$$

Proof. Combine Theorems 5.1 and 6.1 with $Q_t \leq Q_c$. □

Corollary 7.2 (Compact rational form).

$$\frac{25087538451}{100000000000} < Q_t(I_{3322}) \leq Q_c(I_{3322}) \leq \frac{501750771}{20000000000},$$

and the exact width of this printed enclosure is

$$\frac{99}{100000000000} = 9.9 \times 10^{-10}.$$

Three consequences should be distinguished.

- (a) **Rigorous physical bound.** The upper endpoint applies to every commuting-operator realization, not merely finite matrices or a numerical relaxation output.
- (b) **Tensor versus commuting.** For this single Bell functional,

$$0 \leq Q_c(I_{3322}) - Q_t(I_{3322}) < 10^{-9}.$$

This does not prove equality of the models, but it rules out a larger separation detectable by I_{3322} .

- (c) **Exact high-dimensional strategy.** The lower endpoint is achieved by an explicit finite tensor model with rational projectors and a rational Schmidt vector. It rigorously confirms the high-dimensional numerical construction without proving that dimension 499 is necessary or that the global supremum is attained.

The exact lower endpoint rounds to 0.250875384514 at twelve decimal places, matching the displayed numerical limit in the prior high-dimensional literature. The present novelty is the proof-carrying exact enclosure rather than discovery of that decimal estimate.

8 Verification architecture and contract

8.1 Proof objects and integrity

All certificates are canonical, non-executable JSON. Each carries a SHA-256 self-hash computed after removing its hash field, and a release-wide manifest authenticates every distributed file. Pickle and other executable serialization formats are excluded from the trusted path.

8.2 Two verification paths

The primary verifier uses only the Python standard library. For the level-four upper object it:

1. generates the canonical 244-word list and signed D_4 representation;
2. constructs the five exact integer symmetry bases;
3. proves each rational block positive definite by fraction-free Bareiss elimination;
4. reconstructs the full Gram matrix and checks (4) coefficient by coefficient;
5. reconstructs all six dimension-499 projectors and evaluates the original probability expression directly.

The second verifier imports no code from the first. It regenerates the symmetry bases in a separate implementation, proves block positivity by exact rational LDL^T elimination, and accumulates the upper identity directly from block columns. It evaluates the lower strategy through the independently coded tridiagonal formula.

All acceptance conditions use explicit exceptions rather than Python `assert`. Continuous integration runs both verifiers under ordinary and optimized Python. A deliberate proof-object corruption is required to fail under `python -O`. Mutation tests alter hashes, Gram data, level-four blocks, projector parameters, parameter indices, dense involutions, and states; every mutation must be rejected.

8.3 Formal checker contract

A level-four upper certificate is accepted only if:

1. its self-hash, canonical word list, dimensions, block ordering, and symmetry convention are exact;
2. all five rational blocks are symmetric and strictly positive definite;
3. deterministic reconstruction yields the claimed D_4 -invariant Gram matrix;
4. every normal-form coefficient of $\beta\mathcal{K} - \mathcal{B}_{3322} - W_4^*YW_4$ is exactly zero;
5. $(\beta - 4)/4$ equals the stored rational upper endpoint.

The dimension-499 lower certificate is accepted only if:

1. the indices $1, \dots, 498$ occur exactly once and every $c_i^2 + s_i^2 = 1$ holds exactly;
2. all six reconstructed matrices are symmetric projectors;
3. the Schmidt vector is nonzero;
4. direct evaluation of (1) agrees exactly with (10) and the stored rational value;
5. exact cross multiplication proves the displayed strict lower target.

Stored decimal strings, numerical eigenvalues, solver statuses, and generator output are never accepted as evidence.

8.4 Theorem-dependency and trust map

Upper branch: canonical group normal form + rational D_4 block data $\rightarrow$ exact block positivity + exact coefficient identity $\rightarrow$ Theorem 5.1.

Lower branch: rational circle parameters + rational Schmidt vector $\rightarrow$ exact projectors + direct probability contraction = tridiagonal contraction $\rightarrow$ Theorem 6.1.

Combination: Theorems 5.1 and 6.1 + the model inclusion $Q_t \leq Q_c \rightarrow$ Theorem 7.1.

Not trusted: SDP solvers, numerical eigensolvers, see-saw searches, rational-reconstruction scripts, stored decimals, or the execution transcripts.

9 Transfer beyond I_{3322}

The verification method is not tied to the numerical value of one inequality. Let $\mathcal{A}$ be a finitely presented $*$ -algebra encoding a Bell or noncommutative polynomial model, let $p = p^* \in \mathbb{Q}\mathcal{A}$, and let W be a finite rational word vector.

Proposition 9.1 (Finite proof-object transfer principle). *Suppose a proof object contains rational data satisfying*

$$\beta\mathcal{K} - p = W^*YW, \quad Y \succeq 0,$$

with equality decidable in a canonical normal form for $\mathcal{A}$. Then exact rational verification proves the corresponding commuting-operator upper bound. If a second proof object contains rational finite-dimensional operators satisfying all defining relations and a nonzero rational state, exact contraction gives a tensor lower bound.

Proof. The upper statement is the same representation-positivity argument as Proposition 3.1; the lower statement is the variational principle. Finiteness and canonical normal form make every acceptance condition decidable by exact arithmetic. $\square$

In practice, transfer requires four engineering ingredients: a canonical word reducer, an exact coefficient collector, a positivity certificate suited to the matrix structure, and an exact model checker for lower strategies. Finite symmetry groups can reduce Gram matrices into rational isotypic blocks exactly as in Section 5. Numerical optimization remains useful for discovery, but exact affine projection and exact positivity are required before publication of the resulting bound.

10 Discussion and limitations

The new level-four certificate reduces the exact uncertainty by about a factor of 265 relative to the preceding level-three release. The tensor and commuting values of I_{3322} are now indistinguishable at the 10^{-9} scale, while the lower point is represented by a concrete dimension-499 strategy rather than an abstract limiting argument.

The remaining questions are structurally harder than producing additional digits:

- *Exact optimality.* The interval does not identify a closed-form optimum.
- *Qubit optimality.* The release does not prove that the qubit value is exactly $1/4$.
- *Finite-dimensional attainment.* A sequence of strong finite strategies does not prove either attainment or non-attainment of the supremum.
- *Tensor-commuting separation.* A narrow common enclosure does not decide whether $Q_t(I_{3322}) = Q_c(I_{3322})$.
- *Dimension witnessing.* The dimension-12 strategy exceeds $1/4$, but a rigorous witness requires matching upper bounds for all smaller allowed dimensions.

These limitations are stated to prevent the exact enclosure from being interpreted as a solution to distinct open problems.

Data and code availability

The accompanying release contains the manuscript source and PDF, canonical JSON certificates, two exact verifiers, optimized-mode and mutation tests, continuous-integration configuration, documentation, exact result tables, untrusted generator material, and a SHA-256 manifest. The principal commands are

```
python3 verifier/verify_all.py
python3 verifier/verify_independent.py
python3 tests/test_mutations.py
bash tests/test_optimized_pristine.sh
bash tests/test_optimized_corruption.sh
```

The command `make verify` runs the primary trusted path. The independent verifier, mutation suite, and optimized-mode guards are available separately through `make second`, `make mutations`, and `make optimized`.

Acknowledgments and disclosure

AI-assisted drafting, code review, and exploratory computation were used during development. All mathematical claims accepted by the release are recomputed by exact verification programs, and the author accepts responsibility for the manuscript and software artifact.

References

- [1] B. S. Cirel'son, Quantum generalizations of Bell's inequality, *Lett. Math. Phys.* **4** (1980), 93–100.
- [2] D. Collins and N. Gisin, A relevant two-qubit Bell inequality inequivalent to the CHSH inequality, *J. Phys. A* **37** (2004), 1775–1787.
- [3] M. Navascués, S. Pironio, and A. Acín, Bounding the set of quantum correlations, *Phys. Rev. Lett.* **98** (2007), 010401.
- [4] M. Navascués, S. Pironio, and A. Acín, A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations, *New J. Phys.* **10** (2008), 073013.
- [5] A. C. Doherty, Y.-C. Liang, B. Toner, and S. Wehner, The quantum moment problem and bounds on entangled multi-prover games, in *Proc. 23rd IEEE CCC* (2008), 199–210.
- [6] K. F. Pál and T. Vértesi, Maximal violation of the I_{3322} inequality using infinite-dimensional quantum systems, *Phys. Rev. A* **82** (2010), 022116.
- [7] H. Peyrl and P. A. Parrilo, Computing sum of squares decompositions with rational coefficients, *Theor. Comput. Sci.* **409** (2008), 269–281.
- [8] W. Slofstra, The set of quantum correlations is not closed, *Forum Math. Pi* **7** (2019), e1.
- [9] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $MIP^* = RE$, arXiv:2001.04383 (2020).
- [10] A. Tavakoli, A. Pozas-Kerstjens, P. Brown, and M. Araújo, Semidefinite programming relaxations for quantum correlations, arXiv:2307.02551 (2023).

A Exact result progression

Object	Exact certified value	Decimal
Bare length-one optimum	3/8	0.375
1+AB upper	1105981489019/4398046511104	0.2514710761304
Full length-two upper	1103645539239/4398046511104	0.2509399426433
Full length-three upper	68960172491/274877906944	0.250875646055647
Full length-four upper	501750771/2000000000	0.2508753855
Dimension-12 lower	exact rational in release	0.250064365190624
Dimension-16 lower	exact rational in release	0.250443306976014
Dimension-499 lower	exact rational in release	0.250875384513977

B Acceptance versus discovery

The level-four blocks were proposed through numerical semidefinite optimization in a symmetry-reduced coordinate system. They were then rationalized, projected onto the exact coefficient constraints, and averaged over the exact signed D_4 action. Those steps are untrusted: a bug can only cause the final exact verifier to reject or, if the output happens to be valid, to accept a mathematically valid certificate. The trusted conclusion depends solely on the exact proof object and checker contract of Section 8.3.